Quotient-space based learning for motion planning infeasibility

Introduction

Motion planning is a fundamental problem in robotics, where the goal is to find a collision-free path that moves a robot from a start configuration to a goal configuration. A complete motion planner either finds such a collision-free path or correctly reports that no valid path exists. In practice, however, achieving completeness is extremely challenging, especially in high-dimensional configuration spaces. As a result, sampling-based motion planners [1, 2], which can quickly explore high-dimensional spaces, are typically employed. These planners are probabilistically complete [3]: if a valid path exists, they are guaranteed to find it given unlimited time. But if no valid path exists, they continue searching until timeout. Therefore, when a planner times out, it does not necessarily mean that the problem is infeasible.

This thesis focuses on the less explored yet important problem of detecting when a motion-planning solution does not exist. Motion infeasibility plays a crucial role in several robotics domains, including Task and Motion Planning (TAMP) [4], rearrangement planning [5], and Navigation Among Movable Obstacles (NAMO) [6].

The main goal of this work is to develop and analyze algorithms that can certify or prove motion-planning infeasibility. To achieve this, the thesis will make use of quotient-space decomposition [6, 7], a technique that breaks a high-dimensional configuration space into a sequence of lower-dimensional spaces. Importantly, proofs of infeasibility established in these lower-dimensional spaces remain valid when extended to the full robot configuration space.

Objectives

Proving infeasibility in high-dimensional configuration spaces is extremely challenging. Existing state-of-the-art methods are limited to spaces of about five dimensions [9, 10], which is far below the dimensionality of many real robot systems. This project investigates how infeasible cases can be detected at lower-dimensional levels instead of reasoning directly in the full configuration space.

The key insight is that, in many manipulation scenarios, infeasibility is often caused by constraints on specific robot links, typically those earlier in the kinematic chain. This means that full high-dimensional analysis is unnecessary.

Given a robot and its environment, the overall goal of the project is to use learning-based techniques to determine if an m-dimensional reduction of the configuration space is valid, thereby enabling faster and more reliable infeasibility detection. To support this, a dataset can be generated by simulating a large number of scenarios and identifying, for each case, whether the original n-dimensional configuration space can be reduced to a lower m-dimensional space for certifying infeasibility.

In summary, the objectives of this research are:

1. Review the literature on motion planning, infeasibility detection, and dimensionality reduction techniques.
2. Gain a deep understanding of the problem of infeasibility in high-dimensional configuration spaces.
3. Develop the theoretical foundations of a framework for detecting infeasibility through quotient-space or lower-dimensional reasoning.
4. Validate and evaluate the proposed approach through simulation experiments and analysis.

References

[1] L. E. Kavraki, P. Svestka, J.-C. Latombe, M. H. Overmars, Probabilistic roadmaps for path planning in high-dimensional configuration spaces, IEEE Transactions on Robotics and Automation 12 (4) (1996) 566–580.
[2] J. J. Kuffner, S. M. LaValle, Rrt-connect: An efficient approach to single query path planning, in: Robotics and Automation, 2000. Proceedings.ICRA'00. IEEE International Conference on, Vol. 2, IEEE, 2000, pp. 995–1001.
[3] S. Karaman, E. Frazzoli, Sampling-based algorithms for optimal motion planning, The International Journal of Robotics Research 30 (7) (2011) 846–894.
[4] Caelan Reed Garrett, Rohan Chitnis, Rachel Holladay, Beomjoon Kim, Tom Silver, Leslie Pack Kaelbling, and Tomás Lozano-Pérez. Integrated task and motion planning. Annual review of control, robotics, and autonomous systems, 4(1):265–293, 2021.
[5] A. Krontiris, K. E. Bekris, Dealing with Difficult Instances of Object Rearrangement, in: Proceedings of Robotics: Science and Systems XI, Rome, Italy, 2015. doi:10.15607/RSS.2015.XI.045.
[6] M. Stilman, J. J. Kuffner, Navigation among movable obstacles: Real-time reasoning in complex environments, International Journal of Humanoid Robotics 2 (04) (2005) 479–503.
[7] A. Orthey, M. Toussaint, Rapidly-exploring quotient-space trees: Motion planning using sequential simplifications, in: The International Symposium of Robotics Research, Springer, 2019, pp. 52–68.
[8] A. Orthey, A. Escande, E. Yoshida, Quotient-space motion planning, in: 2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), IEEE, 2018, pp. 8089–8096.
[9] S. Li, N. T. Dantam, A sampling and learning framework to prove motion planning infeasibility, The International Journal of Robotics Research, 42 (10) (2023) 938–956.
[10] S. Li, N. T. Dantam, Scaling Infeasibility Proofs via Concurrent, Codimension-one, Locally-updated Coxeter Triangulation, IEEE Robotics and Automation Letters 8 (12) (2023) 8303–8310.

Other relevant material

- https://lavalle.pl/planning/ch4.pdf
- https://modernrobotics.northwestern.edu/nu-gm-book-resource/chapter-2-playlist/#department